

1 Original CFG Paper

Tim Salimans Jonathan Ho. “Classifier-Free Diffusion Guidance”. *NeurIPS 2021 Workshop on Deep Generative Models and Downstream Applications* (2021)

Summary

- CFG extends vanilla classifier guidance to trade off mode coverage and sample fidelity in conditional diffusion model sampling in analogy to low temperature sampling or truncation in other types of generative models.
- CFG may be better than vanilla classifier guidance could be seen as an adversarial attack on classifiers, which could boost classifier-based metrics like FID and IS
- Implementing truncation or low temperature sampling in diffusion models are ineffective. Scaling model scores or decreasing the variance of Gaussian noise of the reverse process generates blurry, low quality samples (diffusion models beat GANs paper).
- Vanilla CG adapts the learned score $\epsilon_{\theta}(\mathbf{z}_{\lambda}, \mathbf{c}) \approx -\sigma_{\lambda} \nabla_{\mathbf{z}_{\lambda}} \log p(\mathbf{z}_{\lambda} | \mathbf{c})$ to include the gradient of the log likelihood of an auxiliary classifier model $p_{\theta}(\mathbf{c} | \mathbf{z}_{\lambda})$ as follows:

$$\tilde{\epsilon}_{\theta}(\mathbf{z}_{\lambda}, \mathbf{c}) = \epsilon_{\theta}(\mathbf{z}_{\lambda}, \mathbf{c}) - w \sigma_{\lambda} \nabla_{\mathbf{z}_{\lambda}} \log p_{\theta}(\mathbf{c} | \mathbf{z}_{\lambda}) \approx -\sigma_{\lambda} \nabla_{\mathbf{z}_{\lambda}} [\log p(\mathbf{z}_{\lambda} | \mathbf{c}) + w \log p_{\theta}(\mathbf{c} | \mathbf{z}_{\lambda})]$$

This results in approximate samples from the distribution:

$$\tilde{p}_{\theta}(\mathbf{z}_{\lambda} | \mathbf{c}) \propto p_{\theta}(\mathbf{z}_{\lambda} | \mathbf{c}) p_{\theta}(\mathbf{c} | \mathbf{z}_{\lambda})^w$$

Results on a toy 2D dataset with 3 classes for which the conditional distribution for each class is an isotropic Gaussian show that the conditional upon applying guidance is non-Gaussian. As guidance strength is increased, each conditional places probability mass farther away from other classes and towards directions of high confidence given by logistic regression, and most of the mass becomes concentrated in smaller regions.

- CFG aims to achieve the same modification of the conditional score but without a classifier. It is done by training a conditional and unconditional model with one neural network and trade off both scores during sampling

$$\tilde{\epsilon}_{\theta}(\mathbf{z}_{\lambda}, \mathbf{c}) = (1 + w) \epsilon_{\theta}(\mathbf{z}_{\lambda}, \mathbf{c}) - w \epsilon_{\theta}(\mathbf{z}_{\lambda})$$

Only a relatively small portion of the model capacity of the diffusion model needs to be dedicated to the unconditional generation task in order to produce classifier-free guided scores effective for sample quality.

Questions

- Trade-off between diversity and sample quality (truncation in GAN, temperature sampling in flows, cfg in diffusion). Intuition why these two things are opposing?
- In the 2D toy example, why is the distribution changed in such a way?
- "Furthermore, $\tilde{\epsilon}_\theta$ is constructed from score estimates that are non-conservative vector fields due to the use of unconstrained neural networks, so there in general cannot exist a scalar potential such as a classifier log likelihood for which $\tilde{\epsilon}_\theta$ is the classifier-guided score." What does that mean?
- So basically CFG is trained to mostly be a conditional model and then the conditional score is again heavily amplified during sampling?

2 Boosting Diffusion Models with an Adaptive Momentum Sampler

Xiyu Wang et al. "Boosting Diffusion Models with an Adaptive Momentum Sampler." *IJCAI* (2024)

Summary

- Standard Diffusion model sampling is independent of previous time steps, which can lead to excessive noise and missing high-level components in the generated images.
- By using an ADAM-like sampler and employing a larger momentum that heavily re-uses the update direction of early steps leads to augmented details while still preserving high-level information. This phenomenon aligns with the two stages of learning for likelihood-based models (Rombach et al. 2022), where the first stage focuses on learning high-frequency details, while the second stage focuses on learning conceptual compositions.

Questions

- "IS (...) is often referred to as a method to measure inter-class diversity and is less sensitive to the diversity of the images inside one label (Lucic et al. 2018; Borji 2019). In contrast, FID can detect intra-class mode collapsing as well (Lucic et al. 2018)." Does that mean that the blind spot of both metrics are non diverse intra-class samples which are however not a mode collapse? Is that even desired?
- The authors show a rate (bits/dim) vs distortion (RMSE) curve. What does that mean?

3 Classifier-Free Guidance is a Predictor-Corrector

Arwen Bradley and Preetum Nakkiran. “Classifier-Free Guidance is a Predictor-Corrector”. *arXiv preprint arXiv:2408.09000* (2024)

Summary

- CFG interacts differently with DDIM and DDPM, and neither sampler with CFG generates the gamma powered distribution $p(x | c)^\gamma p(x)^{1-\gamma}$
- CFG can be interpreted as a predictor-corrector methods that alternates between denoising and sharpening
- In the SDE limit, CFG is equivalent to combining a DDIM predictor for the conditional distribution with a Langevin dynamics corrector for the gamma-powered distribution $p_t(x)^{1-\gamma'} p_t(x | c)^{\gamma'}$, with $\gamma' = (2\gamma - 1)$
- Understanding CFG (p^* as the ideal CFG sampler, $\hat{p}$ as the real CFG sampler) - Why CFG improves both image quality and prompt adherence compared to conditional sampling?:

$$\underbrace{\text{PerceivedQuality} [\hat{p}_\gamma]}_{\text{Real CFG}} = \underbrace{\text{PerceivedQuality} [p_\gamma^*]}_{\text{Ideal CFG}} - \underbrace{(\text{PerceivedQuality} [p_\gamma^*] - \text{PerceivedQuality} [\hat{p}_\gamma])}_{\text{Generalization Gap}}$$

Which means at least one of the following occurs:

- The ideal CFG sampler improves in quality with increasing γ . That is, CFG distorts the population distribution in a favorable way (e.g. by sharpening it, or otherwise).
- The generalization gap decreases with increasing γ . That is, CFG has a type of regularization effect, bringing population and empirical processes closer.

It is likely that both occurs: most models are trained on poor-quality datasets (cluttered images from the web) and we want the model to sample from a higher-quality distribution (images of an isolated subject). On the other hand, CFG seems to produce less outliers.

- It is open to find explicit characterizations of CFGs output distribution, in terms of the original $p(x)$ and $p(x|c)$ — although it is possible tractable expressions do not exist.

Questions

- “We cannot mathematically specify this notion of quality, but we will assume it exists for analysis. Notably, PerceivedQuality is not a measurement of how close a distribution is to the ground-truth $p(x=c)$ — it is possible for a generated distribution to appear even “higher quality” than the ground-truth, for example.” How?

- For the case of the reduced generalization gap, is that somehow related to the gap in the ELBO?

4 What does guidance do? A fine-grained analysis in a simple setting

Muthu Chidambaram et al. “What does guidance do? A fine-grained analysis in a simple setting”. *NeurIPS* (2024)

Summary

- The paper analyzes two cases: (1) mixtures of compactly supported distributions and (2) mixtures of Gaussians, which reflect salient properties of guidance that manifest on real-world data.
- As the guidance parameter increases, the guided model samples more heavily from the boundary of the support of the conditional distribution. The guided ODE first swings past the edges of the support of p and into the tails of the noised data distribution p_t before returning. As a result, errors in score estimation at these tails can move the sampling process away from the intended trajectory and thus prevent the trajectory from ever returning to the support of p , leading to corrupted outputs.
- For any nonzero level of score estimation error, sufficiently large guidance will result in sampling away from the support, theoretically justifying the empirical finding that large guidance results in distorted generations.

5 Guiding a Diffusion Model with a Bad Version of Itself

Tero Karras et al. “Guiding a diffusion model with a bad version of itself”. *NeurIPS* (2024)

Summary

- CFG uses an unconditional model to guide a conditional model, leading to simultaneously better prompt alignment and higher-quality images at the cost of reduced variation. Interestingly, it is possible to obtain disentangled control over image quality without compromising the amount of variation by guiding generation using a smaller, less-trained version of the model itself rather than an unconditional model.
- CFG forces the model to generate samples away from other classes, which can mean dropping density areas close to another class. CFG also avoids low-probability intermediate regions which can mean gaps in the density

where CFG wont sample from. The authors say this may stem from a worse fit of the data distribution for the unconditional model due to the fact that it is much more difficult to generate out of the entire distribution, paired with the fact that the unconditional model usually gets a small part of the training budget. This could mean CFG is not only boosting the likelihood of the sample having come from the target class, but also that of having come from the higher-quality implied distribution.

- Autoguidance uses this intuition to guide a high-quality model with a poor quality model, usually with low capacity or under-trained. This is because under limited model capacity, score matching tends to over-emphasize low-probability (i.e., implausible and under-trained) regions of the data distribution. The authors expect the weaker version of the same model to make broadly similar errors in the same regions, only stronger. Measuring the difference between both scores indicates the general direction towards better samples if the models disagree. The authors recommend an early snapshot of a smaller model as the guidance model. Many other degradation models to yield a guidance model are not performing well.

Questions

- "The training objective of a diffusion model aims to cover the entire (conditional) data distribution. This causes problems in low-probability regions: The model gets heavily penalized for not representing them, but it does not have enough data to learn to generate good images corresponding to them. CFG has become the standard method for "lowering the sampling temperature", i.e., focusing the generation on well-learned high-probability regions. By training a denoiser network to operate in both conditional and unconditional setting, the sampling process can be steered away from the unconditional result — in effect, the unconditional generation task specifies a result to avoid.". Should the training objective then not be fixed to ignore low probability regions from the start?

6 Applying Guidance in a Limited Interval Improves Sample and Distribution Quality in Diffusion Models

Tuomas Kynkäänniemi et al. "Applying Guidance in a Limited Interval Improves Sample and Distribution Quality in Diffusion Models". *NeurIPS* (2024)

Summary

- The authors show that guidance is clearly harmful toward the beginning of the backward process (high noise levels) leading towards a handful template images per prompt greatly reducing variation, largely unnecessary

toward the end (low noise levels), and only beneficial in the middle as it causes samples with more decisive set of features with crisper and perceptually more pleasing results.

- Limiting the guidance interval allows the use of much higher guidance weight.
- The proposed guidance interval is based on a piecewise-linear function, which sets w to 1 outside the guidance interval, removing the influence of the unconditional model. Smoother weighing functions did not improve results. Disabling guidance at selected sampling steps also did not work, guidance seems to have a cumulative effect over multiple consecutive steps.

Questions

- The authors claim to estimate the optimal guidance interval in practice, the lower limit should be set to zero first to optimize the upper limit, then fix the upper limit and optimize the lower limit. However, they wrote in the introduction that limiting the guidance interval allows for a higher guidance weight. So it seems counterintuitive that one can have a constant weight but with difference sized intervals.

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